Test of Hypothesis (2)

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Steps (t Test)

$$t = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}}$$

$$Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

1. Identify the hypotheses
2. Choose the value of α; level of significance
3. Appropriate test statistic (Z *or* t) and determine the value.
4. Compare this value with critical value obtained by using α. the critical value depends on the number of degrees of freedom.
5. Decision

Decision: We may reject H_0 if,

a) For left tailed: $t_{cal} \leq -(t_{\alpha,DF})$

b) For right tailed: $t_{cal} \geq +(t_{\alpha,DF})$

c) For two tailed: $|t_{cal}| \geq t_{\frac{\alpha}{2},DF}$

Degrees of Freedom

- Degrees of freedom are the maximum number of logically independent values, which may vary in a data sample.

Degrees of Freedom

- Difference between total sample and total number of parameters used in the test.

$H_0: \mu = \mu_0$

$$t = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}}$$

There is one parameter

Sample size = n

$n - 1$

Degrees of Freedom

Degrees of Freedom

- Difference between total sample and total number of parameters used in the test.

$H_0: \mu_1 = \mu_2$

n_1 n_2

There are two parameter

Sample size $= n_1 + n_2$

$n_1 + n_2$ $-$ 2

Degrees of Freedom

Steps (t Test)

$$t = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}}$$

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1. Identify the hypotheses
2. Choose the value of α; level of significance
3. Appropriate test statistic ($Z\ or\ t$) and determine the value.
4. Compare this value with critical value obtained by using α. the critical value depends on the number of degrees of freedom.
5. Decision

$$df = (n - 1)$$

Decision: We may reject H_0 if,

a) For left tailed: $t_{cal} \leq -(t_{\alpha,DF})$

b) For right tailed: $t_{cal} \geq +(t_{\alpha,DF})$

c) For two tailed: $|t_{cal}| \geq t_{\frac{\alpha}{2},DF}$

Example

$H_0: \mu = 45$

$H_1: \mu \neq 45$

$\alpha = 0.05$

- A fitness coach claims that the average daily exercise time of gym members is 45 minutes. A random sample of 10 gym members recorded their daily exercise times (in minutes) as:43, 47, 50, 45, 42, 44, 48, 46, 49, 43.

$$t_{CAL} = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}}$$

$$t_{CAL} = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}} = \frac{(45.7 - 45)}{\frac{2.75}{\sqrt{10}}} = 0.80$$

$$\bar{X} = \frac{\sum X_i}{n} = 45.7$$

$$s = \sqrt{\frac{\sum (X_i - \bar{X})^2}{n - 1}} = 2.75$$

Example

$H_0: \mu = 45$

$H_1: \mu \neq 45$

$\alpha = 0.05$

$$t_{CAL} = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}} = 0.80$$

- A random sample of 10 boys had the following I. Q's: 70, 120, 110, 101, 88, 83, 95, 98, 107, 100. Do these data support the assumption of a population mean I. Q. of 100?

$t_{TAB} = t\left(\frac{\alpha}{2}, DF\right)$

$t_{TAB} = t(0.025, DF)$

$t_{TAB} = t\big(0.025, (n-1)\big)$

$t_{TAB} = t(0.025,9)$

$t_{TAB} = 2.262$

Decision: We may reject H_0 if,

a) For left tailed: $t_{cal} \leq -(t_{\alpha,DF})$

b) For right tailed: $t_{cal} \geq +(t_{\alpha,DF})$

c) For two tailed: $|t_{cal}| \geq t_{\frac{\alpha}{2},DF}$

$|t_{CAL}| < t_{TAB}$

May not reject H_0

Example

$H_0: \mu$

$H_1: \mu <$

$\alpha = 0.01$

$$t_{CAL} = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}}$$

- A customer service department claims that the average response time to complaints is less than 24 hours. A sample of 12 complaints has an average response time of 25.5 hours with a standard deviation of 3 hours. Using a 99% confidence level, can the claim that the average response time is less than 24 hours be rejected)

Errors in Test

- Any decision we make based on a hypothesis test may be incorrect
- Because we have used partial information obtained from a sample to draw conclusion about the entire population.

Errors in Test

- Two types of error can be made

a. Type I error

b. Type II error

Type I Error α

- The probability of rejecting null hypothesis when it is true.

- An organization is evaluating whether a job training program improves employee productivity. Null Hypothesis is the job training program has no effect on productivity and Alternative Hypothesis is The job training program improves productivity.

- If, in reality, the training has no effect (null hypothesis is true), but the statistical analysis leads the researcher to reject the null hypothesis, it would be a Type I error.

Type II Error β

- The probability of accepting null hypothesis when it is false.

- An organization is evaluating whether a job training program improves employee productivity. Null Hypothesis is the job training program has no effect on productivity and Alternative Hypothesis is The job training program improves productivity.

- If, in reality, the training has effect (null hypothesis is true), but the statistical analysis leads the researcher to accept the null hypothesis, it would be a Type II error.

Type I and II

	Actual situation		
Decision from sample		H_0 is true	H_0 is false
	H_0 accepted		
	H_0 rejected		

Type I and II

	Actual situation		
Decision from sample		**H_0 is true**	**H_0 is false**
	H_0 accepted	**No error**	**Type II error**
	H_0 rejected		

Type I and II

	Actual situation		
Decision from sample		H_0 is true	H_0 is false
	H_0 accepted	No error	Type II error
	H_0 rejected	Type I error	No error

Reduce errors

- We can reduce type II error by increasing sample size.

- We can reduce type I error by decreasing level of significance.

Thank You!